

CYLIAH MOSESI

BRIGHTLEARN

EXERCISE 03 : SQL FOUNDATIONS : CASE STATEMENTS

Question 01: Classify each product by price into the three tiers below. 'Expensive' if price > 1000
'mid-range' if price between 100 and 1000
'Budget' if price < 100.

Answer 01:

```
SELECT product_name, price,  
CASE WHEN price > 1000 THEN 'Expensive'  
WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'  
WHEN price < 100 THEN 'Budget'  
END AS price_category  
FROM products;
```

product-name	price	price-category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

Question 02: Label each order by its value. 'High value' for orders ≥ 1000 'Medium value' for 500-999, 99
'Low Value' for orders < 500

ANSWER 02:

```
SELECT customer_name, amount,  
CASE WHEN amount > 1000 THEN 'High value'  
      WHEN amount BETWEEN 500 AND 999.99  
      THEN 'Medium value'  
      ELSE 'Low-value'  
END AS order_value_category  
FROM orders;
```

customer_name	amount	order_value_category
Alice	150	Low value
Bob	560	Medium value
Charlie	999	Medium value
Diana	45	Low value
Ethan	1200	High value

Question 03: Categorize each employee's position level using both department and salary. IF in 'IT' and salary > 80000 → 'Senior IT'. IF in 'HR' and salary > 55000 → 'Experienced HR'. Otherwise → 'Staff'.

ANSWER 03:

```
SELECT emp_name, department, salary,  
CASE WHEN department = 'IT' AND salary > 80000 THEN  
'Senior IT'  
      WHEN department = 'HR' AND salary > 55000 THEN  
'Experienced HR'  
      ELSE 'Staff'  
END AS position_level  
FROM employees;
```


emp_name	department	salary	position_level
John	IT	85 000	Senior IT
Sara	HR	60 000	Experienced HR
Mark	IT	75 000	Staff
Lucy	Finance	95 000	Staff
Tom	HR	55 000	Staff

Question 04: Assign each student a letter grade based on their score. $\geq 90 \rightarrow 'A'$ $80-89 \rightarrow 'B'$ $70-79 \rightarrow 'C'$ $60-69 \rightarrow 'D'$ $< 60 \rightarrow 'F'$

ANSWER 04:

```

SELECT student_name, score,
CASE WHEN score >= 90 THEN 'A'
      WHEN score BETWEEN 80 AND 89 THEN 'B'
      WHEN score BETWEEN 60 AND 69 THEN 'D'
      WHEN score BETWEEN 70 AND 79 THEN 'C'
      WHEN score < 60 THEN 'F'
END AS grade
FROM students;

```

Student-name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

Question 05: Label delivery performance based on the time taken. ≤ 30 mins $\rightarrow$ 'Fast', $31-60$ mins $\rightarrow$ 'On time', > 60 mins $\rightarrow$ 'Late'.

ANSWER 05:

```
SELECT delivery_id, delivery_time_minutes,  
CASE WHEN delivery_time_minutes < 30 THEN 'Fast'  
WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN  
'on time'  
WHEN delivery_time_minutes > 60 THEN 'LATE'  
END AS performance  
FROM deliveries;
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

Question 06: Convert the numeric priority into a readable label.

ANSWER 06:

```
SELECT issue_type, priority,  
CASE WHEN priority = 3 THEN 'High'  
WHEN priority = 2 THEN 'medium'  
WHEN priority = 1 THEN 'Low'  
END AS priority_label  
FROM tickets;
```

issue type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

Question 07: Calculate attendance percentage and classify the result.

Answer 07.

```
SELECT student_id,  $\frac{\text{days\_present} \times 100}{\text{total\_days}}$  AS percentage,
```

```
CASE WHEN  $\frac{\text{days\_present} \times 100}{\text{total\_days}} \geq 90$  THEN 'Excellent'
```

```
WHEN  $\frac{\text{days\_present} \times 100}{\text{total\_days}}$  BETWEEN 75 AND 89 THEN
```

```
'Good'
```

```
ELSE 'Needs Improvement'
```

```
END AS attendance_status
```

```
FROM attendance;
```

student_id	percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

Question 08: Label the stock of each product

ANSWER 08:

```
SELECT product_id, stock_qty,
```

```
CASE WHEN stock_qty = 0 THEN 'out of Stock'
```

```
WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
```

```
WHEN stock_qty > 5 THEN 'IN stock'
```

```
END AS stock_status
```

```
FROM products_inventory;
```

product_id	stock_qty	Stock_status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

Question 09 : Classify each class by the number of enrolled students.

ANSWER 09

```

SELECT subject, enrolled_students,
CASE WHEN enrolled_students ≥ 25 THEN 'Large'
WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
WHEN enrolled_students < 10 THEN 'Small'
END AS class_size_category
FROM classes;

```

Subject Class-id	Enrolled_students subject	Class-size-category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

Question 10: Apply a discount flag based on the payment method and amount.

ANSWER 10

```
SELECT payment-id, payment-method, amount,  
CASE WHEN payment-method = 'CASH' AND amount ≥ 200  
THEN 'Eligible for discount'  
WHEN ELSE 'Not eligible'  
END AS discount_eligibility  
FROM payments;
```

payment-id	payment-method	amount	discount_eligibility
1	card	50.00	Not eligible
2	cash	200.00	Eligible for discount
3	card	150.00	Not eligible
4	PayPal	75.00	Not eligible
5	Cash	300.00	Eligible for discount.

END